

Elevator Power and Forward CG Limit

In Example 6.1.1 we saw that within the approximations used, the location of the stick-fixed maneuver point is independent of airspeed. We also saw that for a stable aircraft, the elevator deflection required to produce a given normal acceleration increases as the airspeed is decreased. Moreover, as the CG is moved forward, the elevator deflection required for a given maneuver is further increased. At low airspeed with the CG forward of the wing quarter chord, large elevator deflections are required to maneuver the airplane. In Fig. 4.4.3 we saw that large negative elevator deflections are required just to trim the airplane under such conditions. Furthermore, with flaps fully deployed in landing configuration, this condition is further aggravated.

EXAMPLE 6.1.3. (*Suggested by Captain Robert J. Niewoehner, Ph.D., Aerospace Engineering Department, United States Naval Academy*) For the wing-tail combination described in Example 6.1.1 with the CG located 0.70 feet aft of the wing quarter chord, plot the load factor as a function of airspeed at sea level for several fixed values of elevator angles. Overlay these plots on a traditional V - n diagram, which is discussed in Sec. 3.9. In addition to the geometric and aerodynamic properties that are used for Example 6.1.1, assume that the stall limits are fixed by $C_{L_{\max}} = 1.5$ and $C_{L_{\min}} = -0.7$. For the load limits use $n_{\max} = 4$ and $n_{\min} = -2$.

Solution. After applying the definition of weight coefficient from Eq. (6.1.30) and rearranging, Eq. (6.1.34) can be readily solved for the load factor as a function of elevator angle and airspeed,

$$n = \frac{C_{L,\alpha} C_{m,\bar{q}} - C_{L,\bar{q}} C_{m,\alpha}}{C_{m,\alpha} C_W (2V^2/g\bar{c}_w) + C_{L,\alpha} C_{m,\bar{q}} - C_{L,\bar{q}} C_{m,\alpha}} + \frac{C_{m,\alpha} C_{L0} - C_{L,\alpha} C_{m0} - (C_{L,\alpha} C_{m,\delta_e} - C_{L,\delta_e} C_{m,\alpha}) \delta_e}{C_{m,\alpha} C_W (2V^2/g\bar{c}_w) + C_{L,\alpha} C_{m,\bar{q}} - C_{L,\bar{q}} C_{m,\alpha}} \frac{2V^2}{g\bar{c}_w} \quad (6.1.55)$$

where, for this particular airplane at standard sea level,

$$C_W \frac{2V^2}{g\bar{c}_w} = \frac{W}{\frac{1}{2}\rho V^2 S_w} \frac{2V^2}{g\bar{c}_w} = \frac{4W}{\rho g S_w \bar{c}_w} = 143.7$$

Thus we see that with both the CG location and the elevator angle fixed, the load factor is a parabolic function of airspeed. From Examples 4.4.2 and 6.1.1, the following aerodynamic coefficients are independent of CG location:

$$C_{L0} = 0.4075, \quad C_{L,\alpha} = 4.8844, \quad C_{L,\delta_e} = 0.4764$$

With the CG located 0.70 feet aft of the wing quarter chord, the dimensionless CG location is $\hat{x}_{CG} \equiv x_{CG}/\bar{c}_w = 0.1283$, and following Examples 4.4.2 and 6.1.1, we obtain the aerodynamic coefficients that depend on CG location

$$C_{L,\bar{q}} = 3.0236, \quad C_{m0} = -0.0007, \quad C_{m,\alpha} = -0.5959,$$